# Varunprasaath R. Selvaraju, Ph.D.

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## PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in thin-film and optical materials characterization, spectroscopic/ellipsometric workflows, DOE/SPC, and technical documentation. Builds reproducible measurement models and process-property datasets that translate spectroscopy, film quality, and troubleshooting data into engineering decisions.

## CORE COMPETENCIES

Optical Metrology | Spectroscopic Ellipsometry | Thin-Film Characterization | Recipe / Model Development | Measurement Variability Reduction | DOE / SPC | Technical Presentations | Cross-Functional Troubleshooting

## WORK EXPERIENCE

### University of Colorado Boulder

July 2025 - Present

**Postdoctoral Researcher**

- Lead DOE-guided process-development studies for bio-based polymer systems, defining analytical readouts, process windows, reproducibility criteria, and SOP-ready specifications.
- Translate FT-IR, thermal-analysis, spectroscopy, and process data into technical recommendations that support measurement repeatability and material-quality decisions.

### University of Colorado Boulder

July 2021 - July 2025

**Graduate Researcher**

- Designed and characterized 15+ electrochromic material variants using UV-Vis-NIR, FT-IR, spectroelectrochemistry, cyclic voltammetry, ellipsometry-informed optical analysis, NMR, GPC, TGA, DSC, and MS-informed interpretation.
- Built structure-property-processing datasets connecting optical response, stability, purity, and degradation signals to material design and troubleshooting recommendations.

### Georgia Institute of Technology

Aug 2019 - July 2021

**Graduate Researcher**

- Co-led AFOSR-funded thin-film coating process development, linking CVD/surface-preparation conditions to morphology, interface quality, optical performance, and reproducible process controls.

### ETH Zurich

May 2017 - July 2017

**Research Assistant**

- Optimized polymer thin-film deposition with DOE strategies and in-situ QCM-D process monitoring to quantify growth kinetics, coating uniformity, and process repeatability.

### University of Colorado Boulder

2021 - 2025

**Laboratory Operations Manager**

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling while coordinating multiple concurrent materials-development projects.

### Georgia Tech 2019-2021

**Teaching Assistant, Organic Chemistry I & Lab**

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

## KEY PROJECTS

### Optical Metrology and Thin-Film Characterization KLA FaST Match

Mapped optical and electrochemical performance across functional thin-film/material systems using spectroscopy, ellipsometry-informed analysis, and structured troubleshooting.

Spectroscopic ellipsometry, UV-Vis-NIR, FT-IR, thin films, DOE

### AFOSR Defense Thin-Film Coatings DoD Funded

Developed CVD/surface-preparation controls and linked process variables to film morphology, interface quality, and optical/material performance.

CVD, QCM-D, surface preparation, process-property analysis

### Process-Controlled Polymer Systems Current

Defined DOE-guided process windows, measurement readouts, and reproducible specifications for polymer material quality.

DOE, SPC, FT-IR, thermal analysis, SOPs

## EDUCATION

**Ph.D. in Chemistry - University of Colorado Boulder** 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

**B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)** 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

## CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

## TECHNICAL SKILLS

**Optical / Materials Characterization:** Spectroscopic ellipsometry, UV-Vis-NIR, FT-IR, spectroelectrochemistry, cyclic voltammetry, TGA, DSC, NMR, GPC/SEC, QCM-D, MS-informed analysis

**Metrology & Modeling Mindset:** Recipe/model development readiness, measurement variability reduction, method comparison, data interpretation, technical presentations, troubleshooting

**Thin Films & Process Development:** CVD coatings, surface preparation, polymer thin films, electrochromic materials, process windows, DOE/SPC, reproducibility studies

**Communication & Tools:** SOPs, technical reports, customer-ready presentations, Excel, Python/basic, JMP/basic, publication-quality writing

## JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.

3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

## CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

## HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)